

OpenClaw Skills Development

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1. Course Content

Course Overview

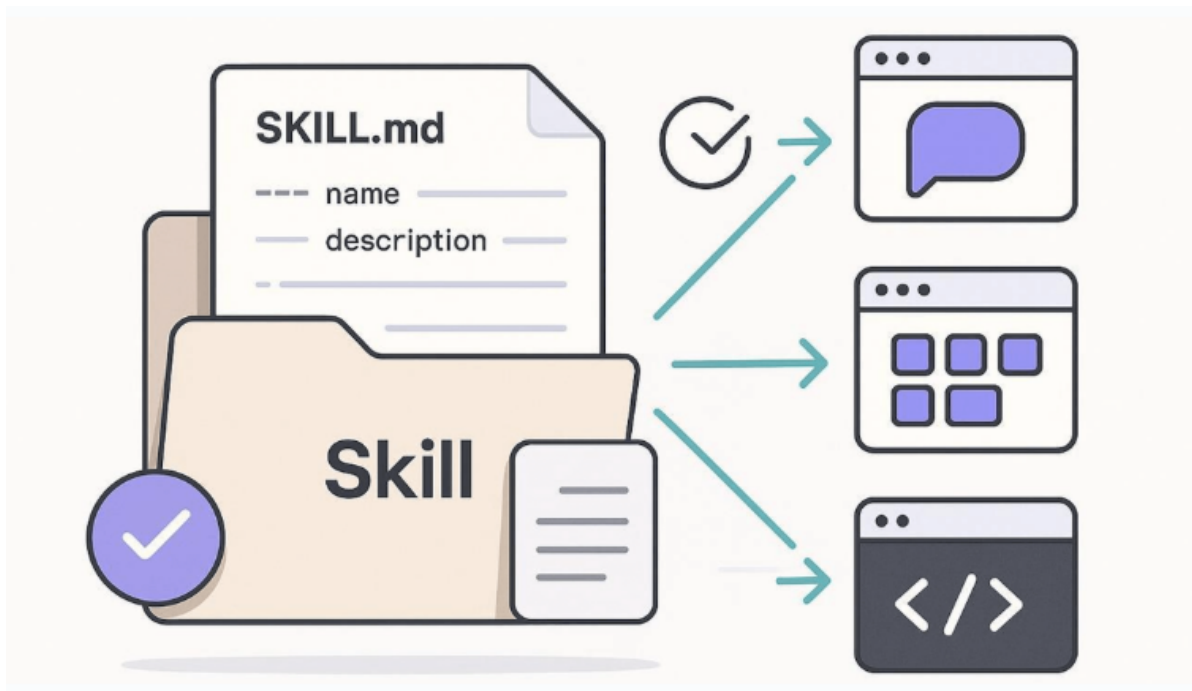
- OpenClaw Skills are a set of advanced task instruction sets for developers. They orchestrate multiple MCP Tools into complete workflows according to business logic, guiding the AI model to call the right tools at the right time to complete complex tasks
- Core skills pre-installed on the robot at the factory: navigation and positioning, visual tracking

Learning Objectives

1. Understand the concept, principles, and functions of OpenClaw Skills
2. Become familiar with the functions and usage of the core skills pre-installed on the robot at the factory
3. Master the operations of enabling and disabling skills in Agents

2. Introduction to Skills

- Skills are a **task-level workflow instruction set**
- Based on the workflow defined in the skill file, the AI model calls MCP Tools step by step to complete the full closed loop from environment perception → decision planning → action execution → result verification
- Skill files are stored in the `$HOME/.openclaw/workspace/skills` directory, with each skill corresponding to a subdirectory whose `SKILL.md` file defines the complete skill instructions



3. Robot Factory-Preset Skills

- You can directly ask OpenClaw in a conversation about the skills it currently has

What skills do you have related to robots?

You 11:41

I have 2 skills related to my robot capabilities:

- 1. **robot-navigation**
 - Navigate to mapped locations on the map
 - Follow line-following (colored lines: red, green, blue, yellow)
 - Record locations with names and symbols
 - Retrieve the map of known locations
- 2. **visual-functions**
 - Observe the environment using my camera
 - Detect and locate target objects (bounding boxes)
 - Track objects visually in real-time

These two skills work together: I can **see** where I am, **find** objects, and **move** toward them or follow paths.

Assistant 11:41

- Skills folder path:

```
$HOME/.openclaw/workspace/skills
```

```
yahboom@yahboom:~$ ls $HOME/.openclaw/workspace/skills
robot-navigation  visual-functions
```

The following 2 core skills are pre-installed on the robot at the factory:

🌀 robot-navigation (Navigation and Positioning)

- **Functions:** Navigate to target locations in the map according to route rules, supporting the recording and reading of map waypoints
- **Core tools:** Navigation / GetMapMapping / RecordMapLocation

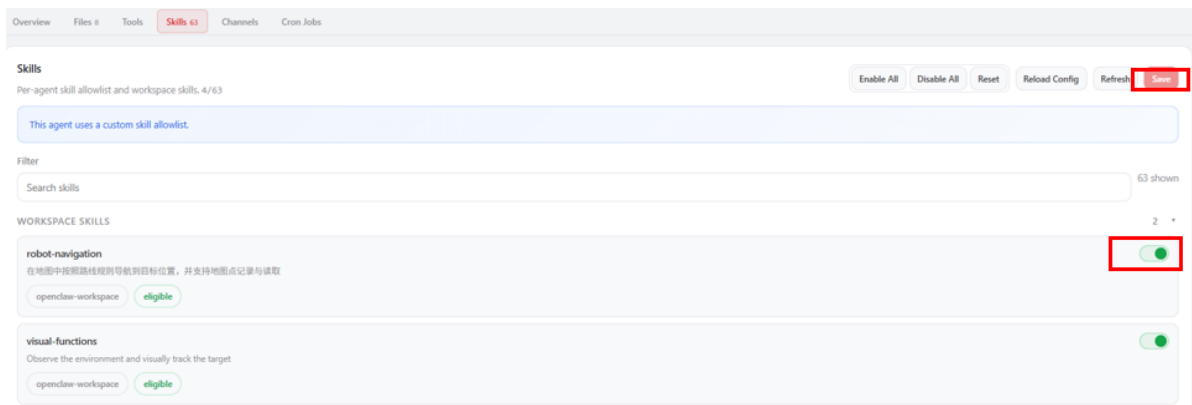
🎯 visual-functions (Vision Functions)

- **Functions:** Observe the environment and perform visual tracking of target objects
- **Core tools:** SeeWhat / GetBbox / TargetTrack

The above skill files are stored in the `$HOME/.openclaw/workspace/skills` directory, one subdirectory per skill. You can directly view or modify the `SKILL.md` files in them

4. Agent Skill Management

- Log in to the OpenClaw WebChat interface, then click **Agent** → **Skills** in sequence
- Under the **Workspace Skills** tab, you can see the factory-preset skills in the workspace. To enable or disable an individual skill, click the switch, then click **Save**



5. Custom Skill Development

5.1 How to Develop Your Own Skill

- Creating a custom Skill is essentially writing a `SKILL.md` file that conforms to the format specification and placing it in the `$HOME/.openclaw/workspace/skills` directory
- Developing a skill requires the following steps:
 1. **Create the skill directory:** Create a new subdirectory under the `skills` directory, with the directory name being the skill name (lowercase English plus hyphens is recommended)
 2. **Write SKILL.md:** Create a `SKILL.md` file in the subdirectory and write the skill content according to the format
 3. **Restart OpenClaw:** Restart the OpenClaw gateway for the new skill to take effect
 4. **Enable in an Agent:** Enable the new skill in the Agent Skills management in WebChat

5.2 Creating a Skill

Create the skill directory and files

```
mkdir -p $HOME/.openclaw/workspace/skills/my-custom-skill
touch $HOME/.openclaw/workspace/skills/my-custom-skill/SKILL.md
```

5.3 Skill Document Format

Each `SKILL.md` file consists of two parts:

YAML metadata header

```
---
name: my-custom-skill           # Skill name (unique identifier)
description: Skill function description # A brief description of the skill's
purpose
version: 1.0.0                 # Version number
author: "Your Name"            # Author information
trigger_keywords:               # Trigger keywords (the AI calls this skill
based on these words)
  - keyword1
  - keyword2
---
```

The body section uses Markdown format to write the skill workflow, usually including:

- **Related tools:** List the MCP Tools required by the skill
- **Rules:** Any necessary constraints and rules
- **Workflow:** Define the operation flow step by step, including specific command-line examples
- **Exception handling:** Disposal plans for common errors

5.4 Writing Skill Content

When writing skill content, follow these principles:

- **Clear step division:** Break the task into clear steps (Step1 / Step2 / ...), each with a clear execution goal and acceptance criteria
- **Complete tool call examples:** For the MCP Tools to be called in each step, provide the complete command format and parameter descriptions
- **Exception handling mechanism:** Provide clear handling strategies for possible error situations (such as target not existing, tool call failure, etc.)
- **Conditional judgment and loops:** Steps that require looping should specify a clear termination condition, and steps that require conditional judgment should give clear judgment criteria

When writing, you can refer to the `SKILL.md` files of the factory-preset skills as templates.
Path:

```
$HOME/.openclaw/workspace/skills/<skill-name>/SKILL.md
```

5.5 Enabling Skills in OpenClaw

- Refer to **4. Agent Skill Management** to enable and disable skills